

Biology · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

122 marks

reading 15 min · writing 150 min

State average on these questions

88 / 122 (72%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the easiest questions.

Sources

- [2022 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2023 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2024 Biology exam \(PDF\)](#) · [examiners' report](#)
- [2025 Biology exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Identifying a single-stranded molecule that contains exons and introns

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 4\)](#) ↗

Identifying the molecules in a diagram of translation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 7 · [open in the official exam PDF \(p. 5\)](#) ↗

An example of a beneficial GM crop trait

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 8 · [open in the official exam PDF \(p. 6\)](#) ↗

Drawing a valid conclusion from a graph of GM crop adoption over time

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 22 · [open in the official exam PDF \(p. 13\)](#) ↗

How immune cells tell self from non-self

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 24 · [open in the official exam PDF \(p. 15\)](#) ↗

What a positive IgM antibody test indicates

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 15\)](#) ↗

Reading the R_0 of measles from an infographic

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2022 VCAA exam · Section A · Question 30 · [open in the official exam PDF \(p. 16\)](#) ↗

Consequences of low genetic diversity in an island koala population founded by a few individuals

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2022 VCAA exam · Section A · Question 32 · [open in the official exam PDF \(p. 17\)](#) ↗

Which findings make fossil formation of an ancient amphibian more likely

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark

2023 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Matching a nucleic acid to its role

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark

2023 VCAA exam · Section A · Question 9 · [open in the official exam PDF \(p. 5\)](#) ↗

Term for wheat that contains genes transferred from another species

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark

2023 VCAA exam · Section A · Question 12 · [open in the official exam PDF \(p. 6\)](#) ↗

Ethical concept behind publishing all results, favourable or not

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark

2023 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 7\)](#) ↗

A true statement about the Calvin cycle

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark

2023 VCAA exam · Section A · Question 20 · [open in the official exam PDF \(p. 9\)](#) ↗

Why glucose use and lactate production are the same at 50 °C regardless of oxygen

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark

2023 VCAA exam · Section A · Question 21 · [open in the official exam PDF \(p. 9\)](#) ↗

Which cells release histamine in an allergic response

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark

2023 VCAA exam · Section A · Question 30 · [open in the official exam PDF \(p. 12\)](#) ↗

The expected genetic result of a selective breeding program in farmed salmon

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark

2023 VCAA exam · Section A · Question 32 · [open in the official exam PDF \(p. 13\)](#) ↗

Most likely explanation for occasional diphtheria outbreaks

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark

2023 VCAA exam · Section A · Question 33 · [open in the official exam PDF \(p. 14\)](#) ↗

Why the Zika outbreak was classed as an epidemic, not a pandemic

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2023 VCAA exam · Section A · Question 37 · [open in the official exam PDF \(p. 16\)](#) ↗

Reading a finch phylogenetic tree for the most recent common ancestor

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2024 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

A true statement about a protein with quaternary structure

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 21 · 1 mark

2024 VCAA exam · Section A · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Structure and role of an RNA molecule in a eukaryotic cell

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 22 · 1 mark

2024 VCAA exam · Section A · Question 6 · [open in the official exam PDF \(p. 4\)](#) ↗

The natural role of CRISPR-Cas9 in bacteria

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 23 · 1 mark

2024 VCAA exam · Section A · Question 10 · [open in the official exam PDF \(p. 7\)](#) ↗

Valid conclusion from high lactic acid in cells

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 24 · 1 mark

2024 VCAA exam · Section A · Question 12 · [open in the official exam PDF \(p. 8\)](#) ↗

A reasonable conclusion from a microalgae growth graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 25 · 1 mark

2024 VCAA exam · Section A · Question 13 · [open in the official exam PDF \(p. 8\)](#) ↗

A controlled variable in a light-colour and algae experiment

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 26 · 1 mark

2024 VCAA exam · Section A · Question 23 · [open in the official exam PDF \(p. 12\)](#) ↗

Which option is an example of primary, quantitative data

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 27 · 1 mark

2024 VCAA exam · Section A · Question 31 · [open in the official exam PDF \(p. 16\)](#) ↗

Reading a fossil diversity diagram of fish groups over time

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 28 · 1 mark

2024 VCAA exam · Section A · Question 34 · [open in the official exam PDF \(p. 18\)](#) ↗

Naming the reduced pelvis bones in modern dolphins

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 29 · 1 mark

2024 VCAA exam · Section A · Question 36 · [open in the official exam PDF \(p. 19\)](#) ↗

Interpreting divergence times on a marsupial phylogenetic tree

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 30 · 1 mark

2025 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Independent variable in a DNA-denaturation experiment

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 31 · 1 mark

2025 VCAA exam · Section A · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Interpreting a haemoglobin activity–temperature graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 32 · 1 mark

2025 VCAA exam · Section A · Question 7 · [open in the official exam PDF \(p. 6\)](#) ↗

The purpose of single guide RNA in CRISPR-Cas9

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 33 · 1 mark

2025 VCAA exam · Section A · Question 12 · [open in the official exam PDF \(p. 9\)](#) ↗

What makes DNA fragments move through an electrophoresis gel

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 34 · 1 mark

2025 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 10\)](#) ↗

Why reduced leaf overlap increases the rate of photosynthesis

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 35 · 1 mark

2025 VCAA exam · Section A · Question 23 · [open in the official exam PDF \(p. 13\)](#) ↗

Identifying the anecdote in a class discussion

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 36 · 1 mark

2025 VCAA exam · Section A · Question 24 · [open in the official exam PDF \(p. 14\)](#) ↗

Matching examples to cellular pathogen, non-cellular pathogen and allergen

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 37 · 1 mark

2025 VCAA exam · Section A · Question 26 · [open in the official exam PDF \(p. 15\)](#) ↗

What happens during an inflammatory response

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 38 · 1 mark

2025 VCAA exam · Section A · Question 31 · [open in the official exam PDF \(p. 16\)](#) ↗

The effect of high vaccination rates in a population

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 39 · 1 mark

2025 VCAA exam · Section A · Question 32 · [open in the official exam PDF \(p. 17\)](#) ↗

Which population factor points to a new rather than re-emerging pathogen

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 40 · 1 mark

2025 VCAA exam · Section A · Question 34 · [open in the official exam PDF \(p. 18\)](#) ↗

Interpreting a dog–dingo–wolf phylogenetic tree

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (7 marks)

2022 VCAA exam · Section B · Question 1 · [open in the official exam PDF \(p. 22\)](#) ↗

Tryptophan and the genetic code: unique codons, transcription/translation from a template, trp operon repression

a. What trp and met share in the genetic code that the other 18 amino acids don't (1 mark)

b. Complete the template, write the mRNA and translate it so that trp is added next (3 marks)

c. Annotate the trp operon to show repression when tryptophan is high (3 marks)

Question 2 (8 marks)

2022 VCAA exam · Section B · Question 3 · [open in the official exam PDF \(p. 26\)](#) ↗

Classifying an unknown plant as C3, C4 or CAM from a data table, then extending the investigation

a. Identify the plant type and justify it from the table (5 marks)

b. Three further steps to help identify the plant (3 marks)

Question 3 (6 marks)

2022 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 32\)](#) ↗

Chickenpox parties vs vaccination; mild disease after vaccination; chickenpox and the Aboriginal population after colonisation

a. Type of immunity gained from being infected at a chickenpox party (1 mark)

b. Why vaccinated people who get infected usually have mild disease (2 marks)

c. Why chickenpox devastated the Aboriginal population but not the colonists, and why numbers recovered 1790–1810 (3 marks)

Question 4 (6 marks)

2022 VCAA exam · Section B · Question 9 · [open in the official exam PDF \(p. 36\)](#) ↗

Hominin fossil ranges: trends in brain and limbs, possible interbreeding, limits of the evidence

Malate dehydrogenase with and without a competitive inhibitor: graphing, mechanism and temperature

a. Plot the rate data with labelled axes (3 marks)

b. How competitive inhibitors affect enzyme activity (2 marks)

c. Effect of running the repeat at 20 °C instead of 37 °C, and why (2 marks)

Question 7 (10 marks)

2023 VCAA exam · Section B · Question 3 · [open in the official exam PDF \(p. 24\)](#) ↗

Antibodies against *S. pyogenes*, bioethics of a human challenge trial, vaccine trial data, drug-resistant TB

a. How specific antibodies help during infection and lead to recovery (3 marks)

b. A bioethical principle the challenge study may have breached, with reason (2 marks)

c. Which participant most likely had the placebo, with reasons (3 marks)

d. Two ongoing challenges caused by drug-resistant TB (2 marks)

Question 8 (9 marks)

2024 VCAA exam · Section B · Question 5 · [open in the official exam PDF \(p. 30\)](#) ↗

CRISPR-edited barley with better nitrogen use: reading graphs, failed edits, identifying the edited plant on a gel

a.i. Read chlorophyll and nitrogen values for control and transformed plants from the graphs (1 mark)

a.ii. How the change (chlorophyll or nitrogen) could affect the plant (2 marks)

b. Why some barley seeds may not be transformed (2 marks)

c. Identify the transformed plant from the gel, justified (2 marks)

d. How gel electrophoresis separates DNA fragments (2 marks)

Question 9 (8 marks)

2024 VCAA exam · Section B · Question 8 · [open in the official exam PDF \(p. 36\)](#) ↗

Two groups of a eucalyptus species in different soils: methodology, species evidence, type of speciation

a. Methodology used to collect the samples (1 mark)

b. Evidence that the two groups are separate species (2 marks)

c.i. Type of speciation (1 mark)

c.ii. Explain how genetic differences arose between the two groups (4 marks)

Question 10 (9 marks)

2024 VCAA exam · Section B · Question 10 · [open in the official exam PDF \(p. 42\)](#) ↗

Student investigation: does highlighting herd immunity change vaccination intentions?

a. Analyse the results (3 marks)

b.i. Recommended strategy based on the findings (1 mark)

b.ii. Another strategy to raise vaccination rates (1 mark)

c.i. Why participants were randomly allocated to groups (2 marks)

c.ii. Comment on the validity of the results (2 marks)

Question 11 (4 marks)

2025 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 30\)](#) ↗

Gene-edited rice vs Bt maize: transgenic or GM, and how increased chlorophyll raises yield

a. Classify each crop as transgenic, GM or both, with justification (2 marks)

b. How editing OSHXK1 could increase crop yield (2 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	B	Single-stranded with introns still present means pre-mRNA. Mature mRNA has had its introns spliced out, and DNA is double-stranded.	86% correct A 6% · B 86% · C 2% · D 5%
☐	2	B	Molecule X is a tRNA carrying its amino acid to the ribosome. mRNA is read 5' → 3', and codons are on the mRNA, not the tRNA.	88% correct A 2% · B 88% · C 6% · D 4%
☐	3	C	Resistance to fungal toxins benefits growers and consumers. The other options are harmful or aren't genetic traits.	86% correct A 1% · B 10% · C 86% · D 3%
☐	4	C	Check each option against the actual graph values for that year. Only one reads the data correctly.	89% correct A 4% · B 6% · C 89% · D 1%
☐	5	B	Non-self cells carry different antigens (non-self markers). Size and reproduction rate are irrelevant.	90% correct A 2% · B 90% · C 7% · D 1%
☐	6	D	IgM is the first antibody made in a primary response, so it shows a current or recent infection.	82% correct A 7% · B 1% · C 9% · D 82%
☐	7	D	Just read the value range given for measles (12–18). Measles is one of the most transmissible diseases.	90% correct A 2% · B 2% · C 6% · D 90%
☐	8	C	A small founding population has less variation, so there are fewer alleles to fall back on if selection pressures change. It doesn't automatically have more mutations.	90% correct A 4% · B 2% · C 90% · D 4%
☐	9	B	Hard parts (a bony skeleton) plus rapid burial in sediment-laden rivers favour fossilisation.	85% correct A 2% · B 85% · C 3% · D 9%
☐	10	B	tRNA carries amino acids to the ribosome. mRNA carries the code, rRNA is structural, and DNA is the template.	86% correct A 4% · B 86% · C 3% · D 7%
☐	11	B	An organism carrying genes from another species is transgenic.	82% correct A 14% · B 82% · C 1% · D 2%
☐	12	C	Reporting everything honestly, including unfavourable results, is integrity.	84% correct A 5% · B 8% · C 84% · D 3%
☐	13	A	The Calvin cycle is light-independent: it uses ATP and NADPH from the light-dependent stage. It doesn't need darkness, and glucose is its product, not an input.	85% correct A 85% · B 11% · C 2% · D 2%
☐	14	A	At 50 °C most glycolytic enzymes have denatured, so the pathway runs at the same low rate whatever the oxygen level.	91% correct A 91% · B 4% · C 3% · D 3%
☐	15	A	Mast cells degranulate and release histamine. Eosinophils, macrophages and NK cells do other jobs.	91% correct A 91% · B 3% · C 4% · D 2%

✓	Q	Ans	Why	How Victoria went
☐	16	C	Humans are the selecting agent, so alleles for the chosen traits increase in frequency each generation.	90% correct A 1% · B 5% · C 90% · D 4%
☐	17	B	Diphtheria is bacterial and vaccine-preventable. Outbreaks follow a loss of herd immunity when vaccination rates drop. Antivirals don't work on bacteria.	83% correct A 7% · B 83% · C 3% · D 8%
☐	18	A	An epidemic is confined to particular regions. A pandemic spreads across many countries or continents.	89% correct A 89% · B 2% · C 9% · D 0%
☐	19	D	Find the pair whose branches join at the node closest to the tips: the small and medium ground finches.	89% correct A 6% · B 3% · C 2% · D 89%
☐	20	A	Quaternary structure means a protein, and proteins are polymers of amino acids. Every protein has a primary structure, and translation happens at ribosomes, not in the nucleus.	90% correct A 90% · B 5% · C 3% · D 2%
☐	21	D	RNA nucleotides contain ribose, and mRNA carries protein instructions to the ribosome. RNA is single-stranded and has uracil, not thymine.	84% correct A 2% · B 10% · C 3% · D 84%
☐	22	B	In bacteria, CRISPR stores spacer sequences from past viruses so Cas9 can cut them if they reinfect. It works like adaptive immunity.	84% correct A 5% · B 84% · C 4% · D 7%
☐	23	D	Lactic acid comes from anaerobic fermentation in animal cells, which happens when oxygen is absent or low. It occurs in the cytosol and produces no CO ₂ .	85% correct A 2% · B 9% · C 4% · D 85%
☐	24	D	Only "white light increased every day" matches the plotted points. Check each claim against every data point.	86% correct A 9% · B 2% · C 2% · D 86%
☐	25	C	Light intensity must be kept the same for all colours. The outputs (O ₂ , glucose) are dependent variables.	84% correct A 2% · B 2% · C 84% · D 12%
☐	26	D	Measuring a mass yourself is both primary (you collected it) and quantitative (numerical). Other students' data or articles are secondary, and smell is qualitative.	88% correct A 6% · B 5% · C 2% · D 88%
☐	27	C	The jawless fish branch starts before 485 million years ago. The other claims aren't supported by the diagram.	84% correct A 3% · B 11% · C 84% · D 2%
☐	28	B	Structures reduced over time and no longer functional are vestigial.	85% correct A 4% · B 85% · C 2% · D 9%
☐	29	A	Groups that split longer ago have had more time to accumulate mutations, so they differ more in DNA sequence.	89% correct A 89% · B 3% · C 7% · D 2%
☐	30	D	The independent variable is what the experimenter changes, here the temperature.	87% correct A 7% · B 2% · C 3% · D 87%
☐	31	C	Activity drops between 30 and 40 °C as the protein denatures, which disrupts the tertiary and quaternary structure. Heat never changes the primary sequence.	88% correct A 5% · B 3% · C 88% · D 4%

✓	Q	Ans	Why	How Victoria went
☐	32	C	sgRNA base-pairs with the target sequence and guides Cas9 to it. Cas9 does the cutting.	90% correct A 6% · B 3% · C 90% · D 2%
☐	33	B	The phosphate backbone makes DNA negatively charged, so it moves towards the positive electrode.	87% correct A 3% · B 87% · C 7% · D 2%
☐	34	D	Less shading means more light reaches the grana (thylakoids) for the light-dependent stage.	85% correct A 10% · B 2% · C 2% · D 85%
☐	35	D	An anecdote is a personal account, not systematic evidence, like Student 4's story about her mother.	85% correct A 3% · B 3% · C 9% · D 85%
☐	36	B	Bacteria and fungi are cellular, viruses and prions are non-cellular, and dog hair and pollen are allergens.	88% correct A 5% · B 88% · C 2% · D 6%
☐	37	D	Vasodilation and more leaky capillaries increase blood flow to the site, so more leukocytes arrive. Natural killer cells don't engulf pathogens (phagocytes do).	94% correct A 3% · B 1% · C 1% · D 94%
☐	38	A	Herd immunity: fewer susceptible hosts means less transmission, which protects people who can't be vaccinated. Vaccination is active, not passive, immunity.	90% correct A 90% · B 6% · C 4% · D 0%
☐	39	C	Travel brings in pathogens the population has never met, so a new pathogen emerges. Falling immunisation rates cause re-emergence.	84% correct A 5% · B 7% · C 84% · D 4%
☐	40	B	Great Danes and Boxers share a more recent ancestor than Great Danes and dingoes, so they have fewer DNA differences.	86% correct A 2% · B 86% · C 2% · D 10%

Multiple-choice score: _____ / 40

Section B – Short-answer questions

Question 1 (7 marks) · state average 4.9 / 7 (70%)

2022 VCAA exam · Section B · Question 1

a. What trp and met share in the genetic code that the other 18 amino acids don't

☐ Each is coded by only one codon (not degenerate)

Answer / notes: Each has only one codon (UGG and AUG), so there is no redundancy.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: _____ / 1

b. Complete the template, write the mRNA and translate it so that trp is added next

☐ Next template triplet ACC

☐ Correct mRNA: AUG AAG CCG AGU CAU UGG

☐ Correct amino acids: met–lys–pro–ser–his–trp

Answer / notes: Template ACC gives the codon UGG (trp). The mRNA is complementary to the template, with U instead of T. Consequential marks were given for correct translation of a wrong mRNA.

State average 2.2 / 3 (73%); 58% scored full marks. Mark distribution: 0: 13% · 1: 13% · 2: 16% · 3: 58%

Your mark: _____ / 3

c. Annotate the trp operon to show repression when tryptophan is high

- ☐ Label the promoter and operator upstream of trpE–trpA
- ☐ Tryptophan binds to (activates) the repressor protein
- ☐ The active repressor binds the operator, blocking RNA polymerase, so the genes are not transcribed

Answer / notes: Trp acts as a co-repressor: it binds the repressor, which then sits on the operator and stops RNA polymerase transcribing the structural genes.

State average 2.1 / 3 (70%); 57% scored full marks. Mark distribution: 0: 20% · 1: 13% · 2: 10% · 3: 57%

Your mark: _____ / 3

Question 2 (8 marks) · state average 5.5 / 8 (69%)

2022 VCAA exam · Section B · Question 3

a. Identify the plant type and justify it from the table

- ☐ C4 plant
- ☐ Optimal temperature range matches C4
- ☐ Same CO₂-fixing pathway (C4 + Calvin cycle), which rules out C3
- ☐ Stomata open during the day, which rules out CAM
- ☐ Low–moderate photorespiration and moderate water loss match C4

Answer / notes: C4. Match it on each row and use the rows that rule out C3 (pathway, photorespiration) and CAM (daytime stomata).

State average 4.6 / 5 (92%); 70% scored full marks. Mark distribution: 0: 2% · 1: 1% · 2: 1% · 3: 2% · 4: 24% · 5: 70%

Your mark: _____ / 5

b. Three further steps to help identify the plant

- ☐ Test for Calvin-cycle enzymes / Rubisco in the bundle sheath cells
- ☐ Compare structure or DNA with known plants
- ☐ Increase the sample size / repeat, or vary another controlled variable and measure the response

Answer / notes: Go beyond the table: check where Rubisco is, compare DNA or leaf anatomy, and repeat with more plants. Most answers just repeated tests already done.

State average 0.9 / 3 (30%); 3% scored full marks. Mark distribution: 0: 37% · 1: 41% · 2: 19% · 3: 3%

Your mark: _____ / 3

Question 3 (6 marks) · state average 4.0 / 6 (67%)

2022 VCAA exam · Section B · Question 7

a. Type of immunity gained from being infected at a chickenpox party

- ☐ Natural active immunity

Answer / notes: Natural active immunity.

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

b. Why vaccinated people who get infected usually have mild disease

- ☐ Memory cells are already present from the vaccine
- ☐ They give a rapid, large antibody response that neutralises the virus before severe disease

Answer / notes: Memory cells launch a fast secondary response, so the virus is cleared before it causes much damage.

State average 1.1 / 2 (55%); 27% scored full marks. Mark distribution: 0: 16% · 1: 57% · 2: 27%

Your mark: _____ / 2

- c. Why chickenpox devastated the Aboriginal population but not the colonists, and why numbers recovered 1790–1810
- ☐ Colonists already had immunity from childhood exposure before arriving
 - ☐ Aboriginal people had no previous exposure, so there was no immunity and adults suffered severe or fatal disease
 - ☐ Children infected later had mild disease and became immune, so the population recovered

Answer / notes: No prior exposure meant no immunity, so adults died. Later, children caught it mildly and gained immunity, which is why numbers rose.

State average 2.2 / 3 (73%); 36% scored full marks. Mark distribution: 0: 6% · 1: 11% · 2: 47% · 3: 36%

Your mark: _____ / 3

Question 4 (6 marks) · state average 4.0 / 6 (67%)

2022 VCAA exam · Section B · Question 9

- a. Two trends from the oldest to the most recent hominins (brain, limbs)

- ☐ Brain size (cranial capacity) increases
- ☐ Limbs: legs get longer relative to the arms (more angled femur), suiting bipedalism

Answer / notes: Bigger brains, and longer legs relative to arms. Foramen magnum answers didn't count because the question asked about limbs.

State average 1.3 / 2 (65%); 45% scored full marks. Mark distribution: 0: 17% · 1: 38% · 2: 45%

Your mark: _____ / 2

- b.i. Two other occasions two hominin species could have interbred, from the table

- ☐ One valid pair overlapping in time and place (e.g. *A. anamensis* & *A. afarensis*)
- ☐ A second valid pair (e.g. *H. habilis* & *H. rudolfensis*, or *H. erectus* & *H. rudolfensis*)

Answer / notes: Choose species whose date ranges and locations overlap, e.g. *A. anamensis* and *A. afarensis* in East Africa.

State average 1.7 / 2 (85%); 76% scored full marks. Mark distribution: 0: 7% · 1: 17% · 2: 76%

Your mark: _____ / 2

- b.ii. Limitations of that evidence

- ☐ Locations are broad, so the species may not actually have met (no precise geographic overlap)
- ☐ Fossil record is incomplete / no molecular (DNA) evidence

Answer / notes: Overlap in dates and regions doesn't prove contact, and the fossils are scarce with no DNA to confirm it.

State average 1.0 / 2 (50%); 25% scored full marks. Mark distribution: 0: 29% · 1: 46% · 2: 25%

Your mark: _____ / 2

Question 5 (8 marks) · state average 5.3 / 8 (66%)

2022 VCAA exam · Section B · Question 10

- a.i. Contents of conventional vaccines and how they improve the response on later exposure

- ☐ Dead/inactivated pathogen (or a pathogen protein)
- ☐ Stimulates memory cells, giving a faster and larger antibody response on re-exposure

Answer / notes: Inactivated pathogen or a protein from it, which leaves memory cells for a faster, bigger secondary response.

State average 1.6 / 2 (80%); 63% scored full marks. Mark distribution: 0: 6% · 1: 31% · 2: 63%

Your mark: _____ / 2

- a.ii. How mRNA vaccines differ

- ☐ They contain mRNA rather than the antigen
- ☐ The person's own cells make the antigen

Answer / notes: mRNA vaccines deliver the code, and your cells make the antigen.

State average 1.5 / 2 (75%); 67% scored full marks. Mark distribution: 0: 15% · 1: 19% · 2: 67%

Your mark: _____ / 2

b. Advantages of mRNA cancer vaccines, plus an ethical concept

- ☐ Can be produced quickly
- ☐ Easily changed (change the sequence) for different cancers
- ☐ Antibodies are specific to cancer-cell antigens
- ☐ An ethical concept applied to vaccine development (e.g. justice: cost/access; non-maleficence: trial safety)

Answer / notes: They are fast to make, easy to re-target and specific. Then apply an ethical concept to this context, e.g. justice if personalised vaccines are too expensive for most people.

State average 2.2 / 4 (55%); 18% scored full marks. Mark distribution: 0: 14% · 1: 15% · 2: 25% · 3: 27% · 4: 18%

Your mark: _____ / 4

Question 6 (7 marks) · state average 4.4 / 7 (63%)

2022 VCAA exam · Section B · Question 11

a. Plot the rate data with labelled axes

- ☐ Axes labelled with units (malate added, mmol; rate, mmol/min)
- ☐ Even scales with all points correctly plotted
- ☐ Both lines drawn and identified with a key

Answer / notes: Two curves on even scales with units and a key. The inhibited curve approaches the uninhibited one at high malate.

State average 1.9 / 3 (63%); 32% scored full marks. Mark distribution: 0: 13% · 1: 15% · 2: 40% · 3: 32%

Your mark: _____ / 3

b. How competitive inhibitors affect enzyme activity

- ☐ The inhibitor has a shape complementary to the active site and binds there
- ☐ So it competes with and blocks the substrate, lowering the rate (overcome by more substrate)

Answer / notes: It binds the active site and blocks the substrate, and adding more substrate outcompetes it.

State average 1.3 / 2 (65%); 58% scored full marks. Mark distribution: 0: 27% · 1: 15% · 2: 58%

Your mark: _____ / 2

c. Effect of running the repeat at 20 °C instead of 37 °C, and why

- ☐ Lower rates / less product formed
- ☐ Less kinetic energy, so fewer enzyme–substrate collisions (below the optimum)

Answer / notes: Lower rates, because at 20 °C there are fewer successful collisions. It isn't denatured, just slower.

State average 1.2 / 2 (60%); 46% scored full marks. Mark distribution: 0: 27% · 1: 27% · 2: 46%

Your mark: _____ / 2

Question 7 (10 marks) · state average 6.7 / 10 (67%)

2023 VCAA exam · Section B · Question 3

a. How specific antibodies help during infection and lead to recovery

- ☐ Antibodies bind specifically to the bacterial antigens
- ☐ This causes agglutination / neutralisation / opsonisation / complement activation
- ☐ This increases phagocytosis and removes the pathogen, so the person recovers

Answer / notes: Antibodies bind antigens, then clump, neutralise or tag the bacteria, so phagocytes clear them faster.

State average 1.4 / 3 (47%); 22% scored full marks. Mark distribution: 0: 31% · 1: 20% · 2: 26% · 3: 22%

Your mark: _____ / 3

b. A bioethical principle the challenge study may have breached, with reason

- ☐ Names a principle (e.g. non-maleficence)
- ☐ Explains how the study breached it (deliberately infecting people causes harm)

Answer / notes: Non-maleficence: the researchers deliberately gave healthy people a sore throat.

State average 1.8 / 2 (90%); 80% scored full marks. Mark distribution: 0: 5% · 1: 16% · 2: 80%

Your mark: _____ / 2

c. Which participant most likely had the placebo, with reasons

- ☐ Participant 1
- ☐ Got a sore throat
- ☐ Had the lowest memory cell/antibody levels, meaning no prior immune response (no vaccine)

Answer / notes: Participant 1: sore throat, low memory cells at stage 2, and only medium antibodies later.

State average 2.1 / 3 (70%); 48% scored full marks. Mark distribution: 0: 15% · 1: 11% · 2: 26% · 3: 48%

Your mark: _____ / 3

d. Two ongoing challenges caused by drug-resistant TB

- ☐ One valid challenge (e.g. new drugs take time and money to develop)
- ☐ A second valid challenge (e.g. overprescribing or incomplete courses, or harder outbreak control)

Answer / notes: New antibiotics are slow and costly to develop, and misuse keeps selecting for resistant strains.

State average 1.4 / 2 (70%); 49% scored full marks. Mark distribution: 0: 12% · 1: 39% · 2: 49%

Your mark: _____ / 2

Question 8 (9 marks) · state average 5.5 / 9 (61%)

2024 VCAA exam · Section B · Question 5

a.i. Read chlorophyll and nitrogen values for control and transformed plants from the graphs

- ☐ All four values correct (control 1.0 mg/g and 0.2%; transformed 1.5 mg/g and 0.5%)

Answer / notes: Control: 1.0 mg/g, 0.2%. Transformed: 1.5 mg/g, 0.5%. Use a ruler.

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

Your mark: _____ / 1

a.ii. How the change (chlorophyll or nitrogen) could affect the plant

- ☐ A direct effect (more light absorbed / more amino acids and nucleic acids)
- ☐ A consequence (more photosynthesis or protein synthesis, so a higher yield)

Answer / notes: More chlorophyll means more light absorbed, so more glucose and a higher yield. Or more N means more amino acids, so more protein.

State average 1.2 / 2 (60%); 46% scored full marks. Mark distribution: 0: 30% · 1: 24% · 2: 46%

Your mark: _____ / 2

b. Why some barley seeds may not be transformed

- ☐ Not every cell takes up the CRISPR-Cas9 components
- ☐ Or the edit fails (sgRNA mismatch, Cas9 doesn't cut / no PAM, or the cut is repaired without the substitution)

Answer / notes: Not all cells receive CRISPR-Cas9, and even when they do, Cas9 may not cut or the repair may not produce the change.

State average 0.7 / 2 (35%); 13% scored full marks. Mark distribution: 0: 45% · 1: 42% · 2: 13%

Your mark: _____ / 2

c. Identify the transformed plant from the gel, justified

- ☐ Hv1-C
- ☐ Its DNA was not cut by BstN1, so it shows a larger (single) band that travelled less far

Answer / notes: Hv1-C: uncut DNA gives a bigger fragment nearer the wells.

State average 1.4 / 2 (70%); 45% scored full marks. Mark distribution: 0: 11% · 1: 44% · 2: 45%

Your mark: _____ / 2

d. How gel electrophoresis separates DNA fragments

- ☐ DNA is negatively charged, so it moves towards the positive electrode
- ☐ Smaller fragments move faster and further through the gel

Answer / notes: DNA moves to the positive end, and small fragments travel furthest.

State average 1.3 / 2 (65%); 51% scored full marks. Mark distribution: 0: 22% · 1: 27% · 2: 51%

Your mark: _____ / 2

Question 9 (8 marks) · state average 5.0 / 8 (62%)

2024 VCAA exam · Section B · Question 8

a. Methodology used to collect the samples

- ☐ Fieldwork

Answer / notes: Fieldwork.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

Your mark: _____ / 1

b. Evidence that the two groups are separate species

- ☐ Different DNA / molecular sequences
- ☐ A second piece of evidence: different morphology (leaf thickness), different soil niche, or no interbreeding

Answer / notes: Significant genetic differences plus distinct morphology or niche. Say why each is evidence rather than copying the stem.

State average 1.6 / 2 (80%); 65% scored full marks. Mark distribution: 0: 4% · 1: 31% · 2: 65%

Your mark: _____ / 2

c.i. Type of speciation

- ☐ Sympatric

Answer / notes: Sympatric: same area, different soils.

State average 0.8 / 1 (80%); 79% scored full marks. Mark distribution: 0: 21% · 1: 79%

Your mark: _____ / 1

c.ii. Explain how genetic differences arose between the two groups

- ☐ Initial variation in the population
- ☐ Different selection pressures (soil phosphorus levels)
- ☐ Thin-leaved trees had a selective advantage in high-P soil, so their alleles increased
- ☐ Reduced gene flow between the groups, and different mutations accumulated

Answer / notes: Variation, then different selection pressure (phosphorus), then differential survival and reproduction, while reduced gene flow let the gene pools diverge.

State average 2.2 / 4 (55%); 23% scored full marks. Mark distribution: 0: 13% · 1: 21% · 2: 22% · 3: 20% · 4: 23%

Your mark: _____ / 4

Question 10 (9 marks) · state average 5.8 / 9 (64%)

2024 VCAA exam · Section B · Question 10

a. Analyse the results

- ☐ Group A had the lowest likelihood (~2.5)
- ☐ Group B was in the middle (~3)
- ☐ Group C (herd-immunity message) was highest (~3.5); the trend/comparison is stated

Answer / notes: Quote values: $A \approx 2.5 < B \approx 3 < C \approx 3.5$. Herd-immunity messaging gave the highest intention.

State average 2.7 / 3 (90%); 82% scored full marks. Mark distribution: 0: 4% · 1: 4% · 2: 11% · 3: 82%

Your mark: _____ / 3

b.i. Recommended strategy based on the findings

- ☐ Promote/advertise the herd-immunity benefit of vaccination

Answer / notes: Advertise the herd-immunity (and individual) benefits.

State average 0.9 / 1 (90%); 89% scored full marks. Mark distribution: 0: 12% · 1: 89%

Your mark: _____ / 1

b.ii. Another strategy to raise vaccination rates

- ☐ A different valid strategy (free vaccines, school programs, no-jab-no-play, incentives)

Answer / notes: Free or subsidised vaccines, or school programs. It has to be different from the findings-based one.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: _____ / 1

c.i. Why participants were randomly allocated to groups

- ☐ Reduces bias by spreading pre-existing beliefs evenly across groups
- ☐ So the information received is the only difference between groups (increasing validity)

Answer / notes: Randomisation spreads prior opinions evenly, so any difference comes from the message alone.

State average 0.8 / 2 (40%); 5% scored full marks. Mark distribution: 0: 32% · 1: 63% · 2: 5%

Your mark: _____ / 2

c.ii. Comment on the validity of the results

- ☐ A valid point for validity (large sample, anonymous responses, Group A as a control, only the message changed)
- ☐ A limitation or a clear judgement (e.g. self-reported intention may not match behaviour)

Answer / notes: Group A acts as a control and responses were anonymous, but stated intention isn't the same as actually getting vaccinated.

State average 0.9 / 2 (45%); 17% scored full marks. Mark distribution: 0: 28% · 1: 55% · 2: 17%

Your mark: _____ / 2

Question 11 (4 marks) · state average 2.5 / 4 (64%)

2025 VCAA exam · Section B · Question 4

a. Classify each crop as transgenic, GM or both, with justification

- ☐ Rice: GM only, because its own gene was edited with no gene from another species
- ☐ Maize: both, because it has genes from another species (Bt), and all transgenic organisms are GM

Answer / notes: Rice is GM (edited, nothing foreign added). Maize is both (Bt genes inserted).

State average 1.3 / 2 (64%); 45% scored full marks. Mark distribution: 0: 15% · 1: 41% · 2: 45%

Your mark: _____ / 2

b. How editing OsHXX1 could increase crop yield

- ☐ More chlorophyll absorbs more light energy
- ☐ So the photosynthesis rate is higher and more glucose is made for growth

Answer / notes: More chlorophyll means more light absorbed, so more photosynthesis and more glucose. Just restating "more chlorophyll" scored zero.

State average 1.2 / 2 (62%); 47% scored full marks. Mark distribution: 0: 22% · 1: 31% · 2: 47%

Your mark: _____ / 2

Your result

Total: _____ / 122 State average on these questions: **88 / 122 (72%)**

Topics to revise:

Questions and texts © Victorian Curriculum and Assessment Authority (VCAA), linked to vcaa.vic.edu.au. Descriptions, marking notes and answers here are paraphrased study notes, not official VCAA marking schemes. State results are from the VCAA examiners' reports.